Apply filters to SQL queries

Project description

A potential security incident occurred after business hours. The objective is to use SQL to retrieve data about a potential security risk after discovering some potential risks involving login attempts and employee machines. Some security updates are also necessary for specific employees' machines.

Retrieve after hours failed login attempts

The AND operator was used to retrieve data about failed log in attempts after 6PM. Using AND returned data that satisfied both conditions that the log in time was after (greater than) 6PM (18:00) and that it failed (success = 0).

```
MariaDB [organization]> select*
    -> from log_in_attempts
    -> where login_time > '18:00' and success = 0;
```

Retrieve login attempts on specific dates

A suspicious event occurred on 2022-05-09. To investigate this event, a review of all login attempts on this day and the day before are needed.

The OR operator was used to retrieve data about login attempts that occurred either on the date of '2022-05-09' or '2022-05-08'. Data was retrieved about both of these dates.

```
MariaDB [organization]> select*
    -> from log_in_attempts
    -> where login_date = '2022-05-09' or login_date = '2022-05-08';
```

Retrieve login attempts outside of Mexico

The team determined that the activity did not originate in Mexico.

To retrieve data about login attempts that did not occur in Mexico, NOT was used to return data in all countries but Mexico. LIKE was also used since Mexico showed up as either "Mexico" and "Mex" in the table so % was also used to allow any word that started with 'mex' but had letters after 'mex' to also return data.

```
MariaDB [organization]> select*
    -> from log_in_attempts
    -> where not country like 'mex%';
```

Retrieve employees in Marketing

The team wants to perform security updates on specific machines in the Marketing department.

```
MariaDB [organization]> select*
    -> from employees;
```

This command was used to retrieve all columns and values from the employees table.

A further search was done to find information about the marketing department that had offices in the east building.

```
MariaDB [organization]> select*
    -> from employees
    -> where department = 'marketing' and office like 'east%';
+-------------+--------------+----------+------------+----------+
| employee_id | device_id    | username | department | office   |
+-------------+--------------+----------+------------+----------+
|        1000 | a320b137c219 | elarson  | Marketing  | East-170 |
|        1052 | a192b174c940 | jdarosa  | Marketing  | East-195 |
|        1075 | x573y883z772 | fbautist | Marketing  | East-267 |
|        1088 | k8651965m233 | rgosh    | Marketing  | East-157 |
|        1103 | NULL         | randerss | Marketing  | East-460 |
|        1156 | a184b775c707 | dellery  | Marketing  | East-417 |
|        1163 | h679i515j339 | cwilliam | Marketing  | East-216 |
+-------------+--------------+----------+------------+----------+
7 rows in set (0.001 sec)
```

Retrieve employees in Finance or Sales

The team needed to perform a different update to the computers of all the employees in the Finance and Sales department. To locate information about these employees, the OR condition was used.

```
MariaDB [organization]> select*
    -> from employees
    -> where department = 'finance' or department = 'sales';
```

Retrieve all employees not in IT

NOT was used to retrieve data from every department that was not Information Technology.

```
MariaDB [organization]> select*
    -> from employees
    -> where not department = 'Information Technology';
```

Summary

Retrieving data from various columns within the employees table was done quickly and efficiently using filters with AND, OR, and NOT. For example, AND was used when needing to find an employee in the marketing department and that also had an office in the east building. This returned results that showed the team which computers needed security updates.